

Benchmark 1 Study Guide

PSYC 4213 Statistics · Modules 1-3 (Chapters 1-3)

Graded Benchmark 1 is proctored in Canvas (Respondus LockDown Browser + Webcam). No notes, textbook, or other aids on the real exam. Use this guide and Methods Market Concept Review to prepare.

What this benchmark covers

- Module 1 - Why we learn statistics (Chapter 1)
- Module 2 - Research design & measurement (Chapter 2)
- Module 3 - Statistical software basics & data handling (Chapter 3)
- Not on Benchmark 1: descriptive statistics, graphs, probability, or hypothesis tests (those come later).

Module 1 - Why statistics?

- Why psychologists and researchers use statistics; everyday vs scientific claims.
- Population vs sample; descriptive vs inferential statistics (intro level).
- Growth mindset about statistics anxiety; statistics as a learnable skill.

Module 2 - Research design & measurement

- Independent vs dependent variables; operational definitions.
- Four scales of measurement (NOIR): nominal, ordinal, interval, ratio - know examples and which summaries are allowed.
- Experimental vs correlational designs; what causal language each supports.
- Validity (internal, external, construct, statistical conclusion) and reliability (test-retest, internal consistency).
- Threats to validity and ways to improve measurement quality.

Module 3 - Software & data handling

- Navigate your software (Jamovi / SPSS / R / Excel / Stata per your preference).
- Enter data, set variable types and measurement levels correctly.
- Import CSV/data files; save project files (.omv, .sav, etc.).
- Variable view vs data view; value labels vs numeric codes.
- Basic workflow: open data -> check types -> run simple descriptives later in the course.

How to study

- Complete Concept Review for Modules 1-3 in Methods Market until objectives are mastered.
- Take the Benchmark 1 practice test (unlimited retakes) for strengths/weaknesses feedback.
- Re-read textbook Ch. 1-3 and Methods Market Topics for modules you miss on practice.
- Offline primary: download this PDF, print the practice packet, enter answers when back online.

Methods Market: <https://methods-market.cleu.com/class/statistics> - Concept Review, practice tests, and Assignment Help for each module.